Saba Abdi

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Tehran - Iran

Reserch Interests

- Applied Operations Research & Analytics
- Queueing Theory & Markov Chain

Education

- **Sharif University of Technology** *Aug. 2022 - Aug. 2026*
  BSc in Industrial Engineering Tehran, Iran
  - GPA: 3.9/4.00 (18.30/20)

Research Experience

- **Bachelor Thesis** *Sep 2025 - Jun 2026*
  supervisor: Prof. Kourosh Eshghi | Sharif University of Technology 18.5/20
  - Developing a Multi-objective location-distribution optimization modeling blood supply chain by considering the effective capacity of Blood Donation Centers, as a way to model blood perishability.
  - Using exponential distribution, the BDC's real capacity to supply blood is determined and the model is solved, allowing for shortage.
  - Solving the problem using epsilon constraint and weighted sum methods.

Teaching Experience

- **Teaching Assistant in Queuing Theory and Probablistic Models** *Spring 2026*
  Prof. Maryam Radman | Sharif University of Technology | IE Department
  - Grading Assignments and providing constructive feedback to students

- **Teaching Assistant in Business Intelligence** *Spring 2026*
  Prof. Shima Soltanzade | Sharif University of Technology | IE Department
  - Held weekly TA sessions to review and clarify key topics covered in lectures.
  - Developed and solved advanced practice problems to help students master problem-solving technique.
  - Designing reinforcing assignments, grading them and providing constructive feedback to students.

- **Teaching Assistant in Fundamentals of Data Mining and its Applications** *Spring 2026*
  Prof. Maryam Bostanara | Sharif University of Technology | IE Department
  - Designed homework assignments to reinforce students' understanding of course material.
  - Graded assignments and provided detailed feedback to support learning progress.

- **Teaching Assistant in Probability Theory and its Applications** *Fall 2025*
  Prof. Nafiseh Sedghi | Sharif University of Technology | IE Department
  - Held weekly TA sessions to review and clarify key topics covered in lectures.
  - Developed and solved advanced practice problems to help students master problem-solving technique.

- **Teaching Assistant in Fundamentals of Data Mining and its Applications** *Fall 2025*
  Prof. Mana Meskar | Sharif University of Technology | IE Department
  - Designed two homework assignments to reinforce students' understanding of course material.
  - Graded assignments and provided detailed feedback to support learning progress.

◦ Maintained direct communication with students to address questions and provide academic guidance.

- **Head Teaching Assistant in Engineering Economics course** *Spring 2025*
  Prof. Maryam Rezapour Niari | *Sharif University of Technology* | *IE Department*
  ◦ Led and managed a team of 11 teaching assistants, tracked progress to maintain quality and consistency.
  ◦ Supervised the course project and provided guidance to team members throughout its design & development.

- **Teaching Assistant in Engineering Economics** *Fall 2024*
  Prof. Maryam Rezapour Niari | *Sharif University of Technology* | *IE Department*
  ◦ Developed comprehensive question to reinforce key concepts.

- **Teaching Assistant in Engineering Statistics** *Fall 2024*
  Prof. Zahra Hasanpour Jesri | *Sharif University of Technology* | *IE Department*
  ◦ Developed comprehensive question & grading assignments.

- **Teaching Assistant in Manufacturing Processes 1** *Fall 2024*
  Prof. Mahmoud Houshmand | *Sharif University of Technology* | *IE Department*
  ◦ Designed & developed homework assignments to reinforce Learning.
  ◦ Grading quizzes & providing constructive feedback to students.

- **Teaching Assistant in Advanced Programming** *Spring 2024*
  Prof. Moslem Habibi & Prof. Majed Nabavian | *Sharif University of Technology* | *IE Department*
  ◦ Created slide materials used by the professors to teach the class.
  ◦ Collaborated within a team to design the course project in two phases: OOP & Full Stack Development

- **Teaching Assistant in Fundamentals of Python Programming** *Fall 2023*
  Mr. Alireza Kadivar | *Sharif University of Technology* | *CE Department*
  ◦ Designed & developed homework assignments to reinforce concepts of python.
  ◦ Maintained direct communication with students to address questions and provide academic guidance.

Job Experience

- **Business Intelligence Analyst - Part Time** *Oct. 2025 -*
  MAF Pars Hypermarkets Company Tehran, Iran
  ◦ MAF Pars Hypermarkets Company is the pioneer of the modern retail industry in Iran, which has succeeded in bringing what is prevalent in the world to Iran with an entrepreneurial mindset.
  ◦ I joined the company after a successful internship. I have contributed to numerous projects as an optimization engineer/BI Analyst; The projects include Sourcing Ordering, Stock tranferring, and root cause analysis.
  ◦ Gaining hands-on experience in the retail sector, enhancing skills in business intelligence, data modeling, and data-driven decision making.

Technical Skills

◦ Python (NumPy, pandas, scikit-learn, OOP concepts)
◦ Microsoft Office Suite (Excel, Access, specialization in Power BI)
◦ SQL server & Tabular Editor
◦ GAMS & Pyomo
◦ minitab
◦ LaTeX
◦ Control Version with GitHub
◦ Web Development (Django & HTML & CSS)

Test Scores

- TOEFL: 117/120 (30, 29, 30, 28)
- GRE: 321, 3.5

Selected Courses

- Fundamental of Python Programming
- Probability Theory and its Applications
- Engineering Statistics
- Advanced Programming
- Operations Research 1
- Operations Research 2
- Decision Theory
- Statistical Quality Control
- Fundamentals of Data Mining and its Applications
- Queuing Theory and Probabilistic Models
- Fundamentals of Business Intelligence
- Fundamentals of Health Systems
- Fundamentals of Supply Chain
- General Economics
- Econometrics
- Lean Manufacturing
- Systems Analysis

Projects

- **Analyzing the root cause of the decrease in some important metrics** *Summer 2026*
  MAF Pars HyperMarkets
  - Conducted root cause analysis of declining business metrics using decomposition techniques and time-series analysis. Analyzed performance trends across different business segments and customer types to identify the key drivers behind metric declines and pinpoint the areas contributing most to the overall performance deterioration.

- **Supply Chain Network Design & Simulation Using AnyLogistix** *Fall 2025*
  SCM Course at IE Department
  - Developed and simulated a three-phase supply chain business model using AnyLogistix. Modeled the end-to-end supply chain, including product importation, factory location and operations, and distribution processes. Conducted supply chain analysis to evaluate network performance and identify key operational insights across different phases.

- **Analyzing the Impact of Student Habits on Academic Performance** *Spring 2025*
  Data Mining Course at IE Department [Link]
  - Collaborating as a team, we cleaned and preprocessed the data, performed EDA, and implemented multiple classification models, KNN, Decision Tree, Random Forest, Naive Bayes, and Logistic Regression. We optimized performance using cross-validation and GridSearchCV, and evaluated models with accuracy, precision, recall, F1-score, and ROC-AUC.

- **Station Placement Optimization & Flow Free Game Optimization** *Fall 2024*
  Operations Research 2 Course at IE Department [Link]
  - Collaborating as a team, we developed and implemented optimization models for two distinct problems, Station Placement and the Flow Free Game, using GAMS and Pyomo. In addition, we conducted sensitivity analyses to evaluate the robustness of the model.

- **Analysis of Employment Data in Iranian Cities** *Fall 2022*
  Fundamentals of Python Programming Course at CE Department [Link]
  - Collaborating as a team, we cleaned datasets, computed key metrics such as employment ratios across demographics and generated usefule insights

Honors and Awards

- **Straight Admission to Msc program at Sharif University of Technology (without Konkoor)** *Spring 2026*
  IE Depatment of Sharif Univeristy of Technology
- **4/4 GPA in 4th & 5th & 6th & 7th & 8th semesters** *Different semesters*
  IE Depatment of Sharif Univeristy of Technology
- **Ranked in the top 5 among over 90 students in different semesters** *2022 - 2025*
  IE Depatment of Sharif Univeristy of Technology
- **Ranked in the top 0.5% over 145,000 high school students in Konkor** *Jun. 2022*
  Iran National Univeristy Exam
  - Accepted into the First Ranked University in Iran, Sharif University of Technology (SUT)

Volunteer Experience

- **Bronze Medal – University Badminton Tournament** *May.2025*
  IE Depatment of Sharif Univeristy of Technology
  - selected as part of an interdisciplinary team representing the Faculties of Industrial Engineering, Materials Engineering, and the Marine College. Competed against five other teams, which represented other faculties, and secured a third-place finish in the annual tournament.
- **Freshman Mentor** *Oct. 2024 - Jun. 2025*
  Sharif University of Technology
  - Chosen as one of nine mentors to help freshmen of IE department.
  - Guided freshmen by offering study strategies and academic advice.
  - Organized meetings to support their academic transition.